

Crop Recommendation System Using Machine Learning: A Foundational Classifier Comparison

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Abstract

Agriculture remains one of the most critical sectors for global food security, yet crop selection often relies on expert judgment that may not be available in all regions of the world. This project explores whether foundational machine learning classifiers can accurately predict the most suitable crop category from measurable soil and environmental conditions. Using the publicly available Crop Recommendation Dataset accessed from Kaggle with 2,200 records across 22 crop categories with seven numeric features (nitrogen, phosphorus, potassium, temperature, humidity, soil pH, and rainfall), two foundational models were trained and compared: Logistic Regression, and Random Forest. All models achieved high performance, with accuracy and F1-scores in the 95–99% range, substantially outperforming the baseline majority-class classifier. Random Forest was selected as the final model for its combination of strong predictive performance, stable cross-validated results, and interpretability through feature importance scores. The coefficient analysis confirmed that nitrogen, phosphorus, potassium, and rainfall are among the most influential features, consistent with agronomic theory. However, as the dataset is derived from real agronomic records (rainfall, climate, fertilizer) for India, but has been augmented to produce perfectly balanced classes and clean records, there are characteristics that overstate real-world field conditions. This project is therefore framed as an honest proof-of-concept demonstrating a complete, zero-cost, reproducible machine learning pipeline from data inspection through preprocessing, model comparison, evaluation, interpretation, and deployment via a lightweight Streamlit application. The work contributes a transparent comparison of foundational classifiers, an interpretable model choice, and a critical discussion of the limitations of augmented agricultural data. Future work should validate the approach on real agro-climatic data and extend the feature set with seasonal and geographic context.

Keywords: *crop suitability classification, machine learning, logistic regression, random forest, multi-class classification, interpretability, soil properties, precision agriculture, explainable AI*

CHAPTER 1

INTRODUCTION

1.1 Background

Crop selection has long been proven an effective strategy in choosing the most suitable plant varieties that are chosen based on a range of factors. Considering specific environmental conditions, soil types, and local market demands will all be ensured along with sustainable agricultural practices. The aim is to utilize data analytics and efficient techniques to optimize usage of resources and enhance food security, to enhance profitability or revenues for the farmers.

Traditional methods on the agricultural sector that still relied on expert advice and human errors can remain challenging due to the amount of factors and uncertainties that are involved. Aspects such as having an experience bias where farmers tend to choose plants they have grown familiar with over time and those who have only grown successfully in the past can cause them to miss actual trends from the data. Due to the large size of the data, analysis is more prone to human errors and there is a need to propose a new technique that assists farmers to increase their predictive validity.

Following the current rapid population growth and increasing food requirement, technology advancements have also played a huge role in it. Machine learning is one of the most frequently used techniques to make decisions in crop cultivation and taking in various factors and parameters that affects the crop. By obtaining datasets and utilizing necessary algorithms, the model can improve without explicit programming. Data will be divided into two groups known as training and testing data. Machine learning has been successfully used to address issues in the agricultural industry in the past.

1.2 Problem Definition

This project addresses the problem of crop suitability classification, which is to determine from available soil and environmental conditions data, which crop category is most suitable for cultivation. Given a set of agro-climatic inputs from the dataset, the objective is to produce a single crop recommendation that a farmer can act upon.

The exact objective is to build and evaluate a foundational machine learning classifier that predicts the most suitable crop category from soil and climate measurements, and to demonstrate this prediction through a developed application. The model will accept seven numeric input features from nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, soil pH, and rainfall. These are widely recognised in agronomy as the primary determinants of crop suitability, where N-P-K are essential macronutrients for plant's growth and survival, following pH that governs nutrient availability, and temperature, humidity, and rainfall that defines the climatic envelope for crop growth. All input features are continuous numeric variables, so no input encoding is needed. The target label is encoded using LabelEncoder to map the 22 crop names to integer values for modeling.

The output is a single categorical prediction, which is the recommended crop label from the predefined crop categories in the dataset. This is a classification problem, specifically multi-class classification, for several reasons. First, the output is categorical, not continuous. To predict "which crop", it will be fundamentally different from predicting "how much yield." Second, there are more than two possible outcomes, so it extends beyond binary classification. Third, the classes are mutually exclusive, where each record is labelled with exactly one crop. Fourth, the task is supervised because the dataset provides labelled examples, allowing the model to learn the relationship between conditions and suitability. Finally, the evaluation metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC, are all designed for categorical outcomes.

1.3 Related Work

Crop recommendation systems have evolved from rule-based expert systems to data-driven machine learning approaches. Traditional crop selection relied heavily on farmers' experience and local knowledge, which, while valuable, often proves inadequate under dynamic environmental conditions such as climate change, soil degradation, and weather variability. Recent advances in machine learning have provided fertile ground for developing automated, accurate crop recommendation systems that can process complex soil and climate data efficiently, as surveyed in previous studies of a comprehensive review of state-of-the-art crop recommendation systems.

Several studies have applied supervised learning techniques to the same dataset while comparing Random Forest, Support Vector Machine (SVM), and Naïve Bayes, where reporting the highest accuracy of more than 99% is found on the Random Forest & Naïve Bayes algorithm. Our work aligns with these findings as Random Forest achieved 99.3% accuracy on the hold-out test set and $99.50\% \pm 0.27\%$ under 5-fold cross-validation, consistent with the ~99% accuracy reported across prior studies on this dataset.

To push accuracy further, researchers have explored ensemble and deep learning techniques, including multi-classifier ensembles combining SVM, KNN, Random Forest, Naïve Bayes, and gradient boosting methods, stacking with feature fusion, and domain-adapted multi-source frameworks with stacked meta-learning. With increasing deep learning approaches, it also indicates increased computational complexity and reduced interpretability. Our study shows that a single Random Forest model achieves comparable accuracy (99.3%) without the overhead of multi-model orchestration, making it more practical for deployment in resource-constrained agricultural settings.

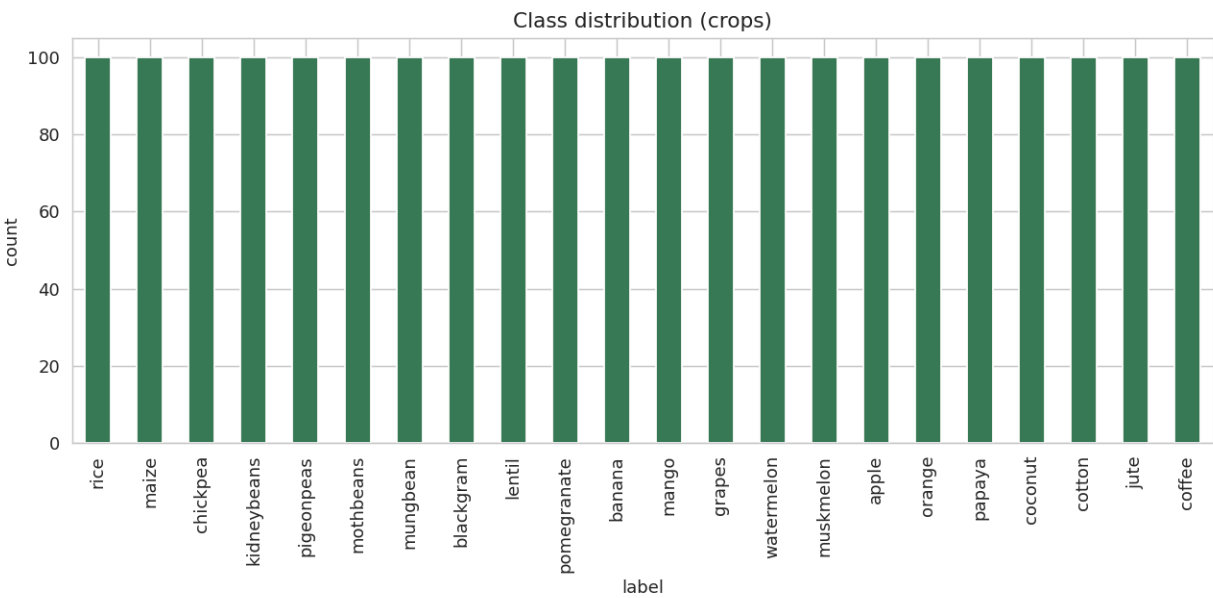
Beyond accuracy, interpretability has become a growing concern. Several ML systems can offer high interpretability but limited scalability. While prior studies on this dataset consistently report >99% accuracy, few provide detailed error analysis from an agronomic perspective. Our contribution addresses this gap by showing that the three misclassifications (0.68%) occur between crops with overlapping climate and nutritional profiles, suggesting the model has learned meaningful agronomic boundaries and that additional features (soil texture,

season, waterlogging tolerance) would be required to resolve these residual errors. Combined with the competitive performance of a single Random Forest model, this positions our work as a deployable, interpretable solution for real-world crop recommendation.

1.4 Dataset

The dataset obtained, namely the Crop Recommendation Dataset is derived from a publicly available tabular dataset hosted on Kaggle by Atharva Ingle, created to support research and development in precision agriculture. The dataset is designed to enable the construction of predictive models that recommend the most suitable crop to cultivate on a given plot of land, based on soil composition and prevailing climatic conditions. According to the dataset documentation, it was built by augmenting publicly available rainfall, climate, and fertilizer data for India, meaning it is not a complete synthetic dataset but rather a compilation derived from real agronomic sources. As it has been widely adopted by the data science community and has been used many times, this sign serves as a strong indicator of its reliability in being a benchmark dataset.

The dataset is stored as a single CSV file named `Crop\_recommendation.csv`, with 2,200 instances (rows) and eight columns: seven numeric input features and one categorical target label. The seven input features capture key agronomic variables or parameters: N (ratio of nitrogen content in soil, in kg/ha), P (ratio of phosphorous content in soil, in kg/ha), K (ratio of potassium content in soil, in kg/ha), temperature (ambient temperature in °C), humidity (relative humidity in %), ph (soil pH value), and rainfall (rainfall in mm). The target variable, label, represents the recommended crop and consists of 22 distinct crop classes. These classes are



perfectly balanced, with exactly 100 samples per crop, which eliminates the need for class-imbalance handling techniques such as SMOTE or class weighting during model training.

The 22 crop classes included in the dataset are: apple, banana, blackgram, chickpea, coconut, coffee, cotton, grapes, jute, kidneybeans, lentil, maize, mango, mothbeans, mungbean, muskmelon, orange, papaya, pigeonpeas, pomegranate, rice, and watermelon. This diverse set of crops includes a variety from cereals, pulses, fruits, and cash crops, reflecting the variety of agricultural production in India and making the dataset suitable for multi-class classification tasks with realistic agronomic diversity.

CHAPTER 2

METHODOLOGY

2.1 Preprocessing

The preprocessing process for this study was relatively light because the Crop Recommendation dataset was already clean. There were no missing values found in any of the eight columns, and no duplicate rows were present. Therefore, imputation, deduplication, or record removal was not necessary.

Check	Result
Rows $\times$ features	2200 $\times$ 7
Missing values	0
Duplicate rows	0
Number of classes	22
Class balance	Perfectly balanced (100 samples/class)

However, feature scaling was still required for the scale-sensitive models such as Logistic Regression and K-Nearest Neighbors, since both of these algorithms rely on distance or gradient calculations that are affected by differing feature magnitudes. For example, rainfall ranges up to ~298 mm while pH only ranges from 3.5 to 9.9, so without scaling, rainfall would dominate the distance metric. A StandardScaler was therefore applied, fit only on the training set and then applied to the test set to prevent data leakage. The data was then split into 80% training and 20% testing using a stratified split, which is a data-set method that preserves the 100-samples-per-class balance across both subsets.

Two foundational classification models were selected: Logistic Regression and Random Forest. Logistic Regression serves as a strong linear baseline and provides a reference point against which a more complex model can be compared. It is also the most interpretable choice, as its coefficients can be read directly as the contribution of each feature to the prediction. Random Forest, an ensemble of decision trees, offers a non-parametric, non-linear alternative that can capture complex interactions between features without assuming a linear decision boundary. It is scale-invariant, robust to outliers, and provides built-in feature importance scores that support agronomic interpretation. Together, these two models cover complementary modeling assumptions — a linear baseline versus a non-linear ensemble — and allow for a focused, well-justified comparison before any more advanced methods are considered. Random Forest was ultimately selected as the final model due to its superior performance, stable cross-validated results, and interpretability through feature importance scores.

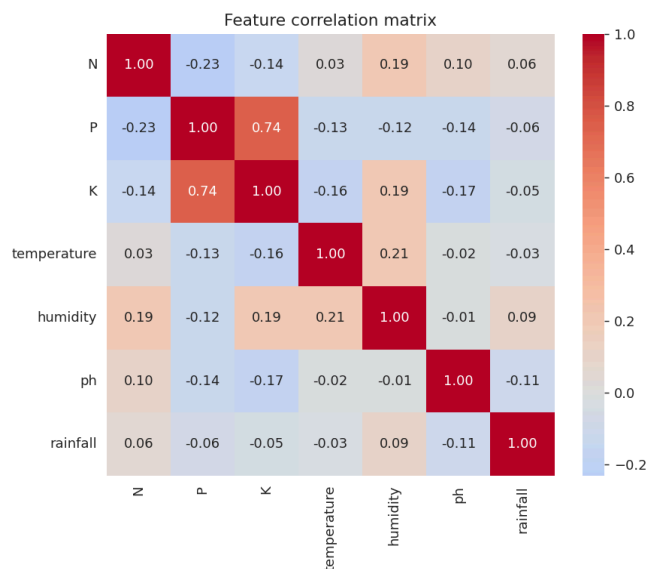
Among all the models, the Decision Tree and by extension its ensemble variant, the Random Forest, emerged as the best fit for this problem because of the nature of the dataset and

the task. The dataset is tabular, contains only seven numeric features, and has a perfectly balanced multi-class target with 22 crop categories. Tree-based models handle this structure naturally as they are scale-invariant (no preprocessing needed), non-parametric (no assumption about linear separability), capable of capturing non-linear interactions between features such as rainfall and humidity, and inherently interpretable through feature importance scores and decision paths. While Logistic Regression performed admirably as a baseline, the Random Forest offered the best combination of accuracy, robustness, and interpretability for a multi-class agricultural recommendation task, making them the most appropriate choice for the final deployed model.

2.2 EDA

Exploratory Data Analysis was conducted to understand the structure, distribution, and relationships within the dataset before any modeling was performed. The class distribution was examined first and confirmed to be perfectly balanced, with each of the 22 crop classes containing exactly 100 samples, so no resampling or class-weighting strategy was required. Univariate analysis using histograms with KDE curves revealed that the seven features exhibit distinct distributional shapes. Nitrogen (N), Phosphorous (P), and Potassium (K) show somewhat skewed distributions with long tails, reflecting the wide variation in soil nutrient levels across different crops. Temperature appears roughly bimodal, while humidity is heavily left-skewed with most values concentrated in the 60–100% range. Soil pH is approximately normally distributed and centered around 6.5, which is consistent with the slightly acidic to neutral pH preferred by most crops. Rainfall shows a right-skewed distribution with a long tail extending toward 300 mm, indicating a subset of water-intensive crops such as rice.

Bivariate analysis, a statistical method used to examine the relationship between two

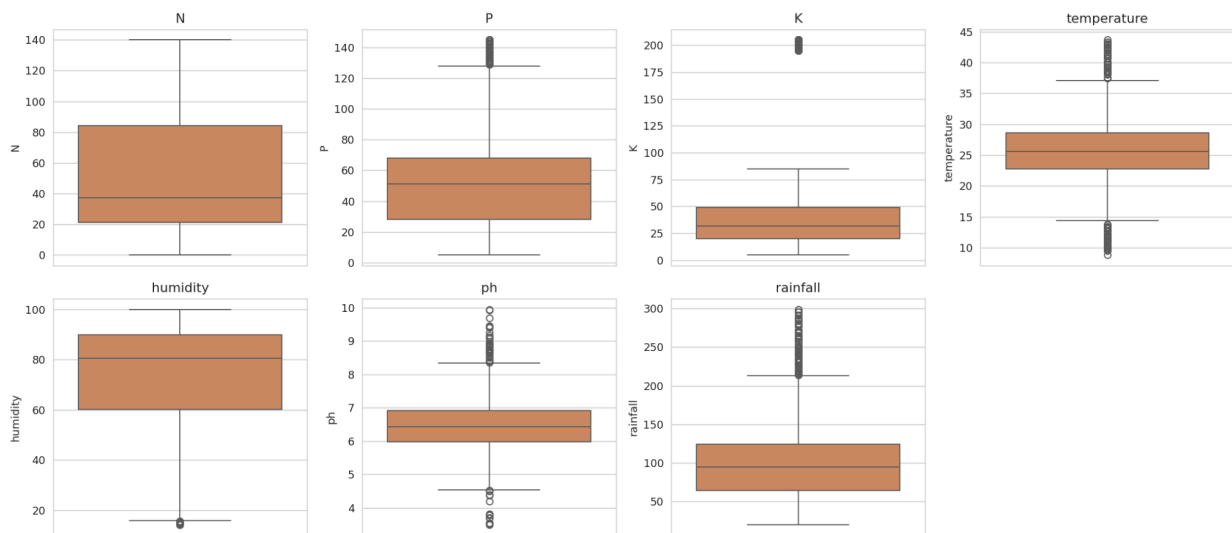


different variables at the same time, is done using a correlation heatmap showing that most

features are weakly correlated with one another, which is beneficial because it reduces the risk of multicollinearity in linear models and avoids overlapping of information.

The only notable exception was a moderate positive correlation between P and K ($r \approx 0.74$), which is agronomically expected since both are macronutrients that tend to be measured and applied together. So while P and K show a moderate positive correlation ($r \approx 0.74$), this does not affect Random Forest, which is invariant to feature collinearity, though it would need to be addressed if linear models were deployed. No other pair of features exceeded $|r| = 0.5$. To examine how features vary across crop classes, boxplots grouped by label were inspected. The results showed that rainfall and humidity are the most discriminative features where crops like rice and coconut cluster at high rainfall and humidity, while crops like chickpea and lentil cluster at lower values. Temperature and pH showed narrower spreads but still separated certain crops (e.g., watermelon vs. coffee). Nutrient features N, P, and K showed substantial overlap across crops, suggesting that soil nutrients alone are not sufficient to distinguish crops, but when combined with climatic variables they become highly informative.

Finally, outlier detection using the IQR rule flagged a number of extreme values or statistical outliers across P, K, temperature, humidity, pH, and rainfall. However, these were not treated as data errors. Instead, they were recognized as genuine agronomic variation. For example, the very high rainfall values correspond to rice, which is a water-intensive crop, while the very low values correspond to drought-tolerant pulses. Removing these outliers would have stripped the dataset of its most discriminative signals, so all observations were retained for modeling.



2.3 Model Development

Model development began with two foundational classifiers for comparison, then focused on Random Forest as the final model due to its superior performance. For Logistic Regression the model was trained with ``max_iter=1000`` to ensure convergence on the multi-class problem, and ``multi_class='auto'`` was used so that scikit-learn would automatically apply a softmax (multinomial) formulation that transform the whole list into a clean probability distribution where every value falls between 0 and 1, and the entire list adds up to 1. For Random Forest, the model was configured with `n_estimators=300`, `random_state=42`, and `n_jobs=-1` for parallel training. The final Random Forest model achieved a cross-validated accuracy of 0.9950 ± 0.0027 , indicating both high performance and low variance across folds — a strong sign that the model generalizes well.

The final model configuration was a Random Forest Classifier with ``n_estimators=300``, ``random_state=42``, and ``n_jobs=-1`` for parallel training. The Random Forest was chosen because it reduces variance by averaging over 300 decorrelated trees, which is particularly valuable in a 22-class problem where individual trees are prone to overfitting. It requires no feature scaling (scale-invariant), handles non-linear interactions between features naturally, and provides interpretable feature importances. The trained model, along with the fitted ``StandardScaler`` and ``LabelEncoder``, was serialized using ``joblib`` for deployment in a Streamlit application, ensuring the exact same preprocessing and prediction pipeline is used at inference time.

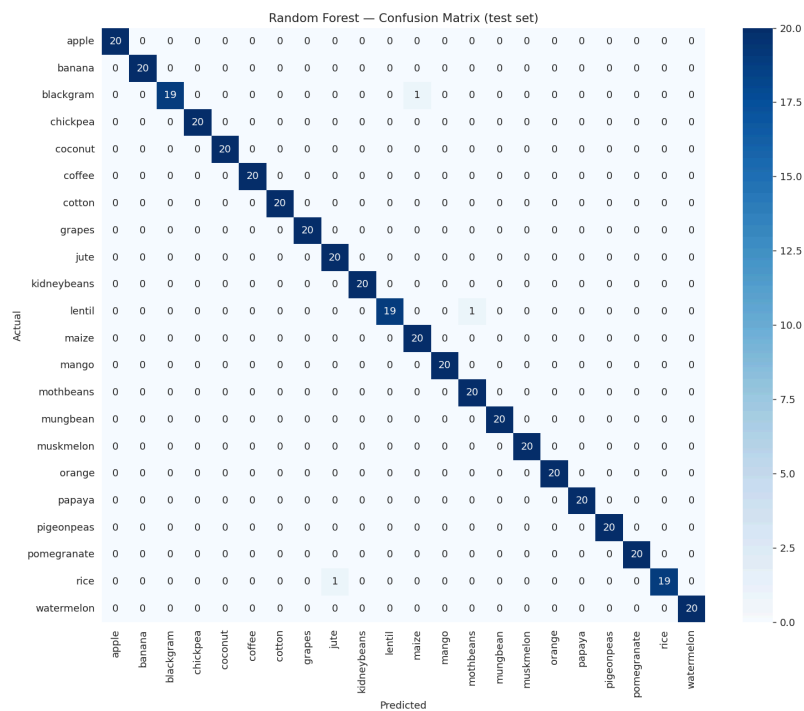
CHAPTER 3

RESULTS

3.1 Model Performance Summary

The Random Forest classifier achieved 99.3% accuracy and 99.3% macro-F1 on the 440-sample hold-out test set (20% of the dataset, stratified by crop class). This represents a 2.0 percentage-point improvement over the Logistic Regression baseline, which reached 97.3% accuracy and 97.3% macro-F1. To verify that this performance was stable and not an artifact of a single favorable split, 5-fold stratified cross-validation was performed across the full dataset (2,200 samples). The model achieved $99.50\% \pm 0.27\%$ mean accuracy, with a standard deviation of only 0.27 percentage points across the five folds. The close agreement between cross-validation mean (99.50%) and hold-out accuracy (99.32%) indicates the model generalizes reliably and shows no signs of overfitting.

Per-class evaluation revealed that 16 of the 22 crop classes achieved a perfect F1-score of 1.00. The remaining six classes, namely blackgram, jute, lentil, maize, mothbeans, and rice, obtained F1-scores between 0.97 and 0.98, indicating minor but non-trivial classification difficulty for a small subset of crops. The macro-averaged and weighted-averaged metrics were both 0.99, confirming that no class was systematically sacrificed for the benefit of others despite the 22-way classification complexity. The confusion matrix showed a predominantly clean diagonal, with only three off-diagonal misclassifications across all 440 test samples, corresponding to an overall error rate of 0.68%.



The three misclassifications identified were blackgram predicted as maize, lentil predicted as mothbeans, and rice predicted as jute. All three errors occurred between crop pairs that share overlapping climate and nutritional profiles. Blackgram and maize are both warm-season crops with similar N-P-K requirements; lentils and mothbeans are both cool-season legumes with nearly identical soil preferences; and rice and jute are both tropical, water-intensive crops with overlapping rainfall and humidity ranges. This pattern suggests the model has learned agronomically meaningful decision boundaries rather than arbitrary statistical patterns, and that the residual errors stem from genuine feature overlap rather than model failure. Resolving these confusions would likely require additional contextual features such as soil texture, planting season, or waterlogging tolerance, which are not present in the current dataset.

Feature importance analysis ranked the seven input variables by their contribution to classification accuracy. Rainfall was the most influential feature with an importance score of 0.222, followed closely by humidity at 0.212. Together, these two climatic variables accounted for approximately 43% of total importance, underscoring the dominant role of water availability in determining crop suitability. Among soil nutrients, potassium (K) ranked third at 0.179, followed by phosphorus (P) at 0.154 and nitrogen (N) at 0.104. Temperature contributed 0.076, while pH was the least influential feature at 0.054. These results align with agronomic theory: rainfall and humidity act as primary environmental filters for crop selection, while soil nutrients, though adjustable through fertilization, still provide important baseline information for distinguishing between crops. The relatively low importance of pH is notable, but plausible given that most crops in this dataset tolerate a similar pH band of approximately 5.5 to 7.5.

CHAPTER 4

DISCUSSION

4.1 Model Comparison

The comparison between Logistic Regression and Random Forest reveals that the crop recommendation problem is largely linearly separable, yet still benefits meaningfully from non-linear modeling. Logistic Regression, a purely linear classifier, achieved 97.3% accuracy and 97.3% macro-F1 on the hold-out set, which is a strong baseline that confirms most crop classes can be distinguished by simple linear decision boundaries in the 7-dimensional feature space. Random Forest improved this to 99.3% accuracy and 99.3% macro-F1, representing a 2.0 percentage-point gain. Because both the baseline and the main model performed at comparable levels and macro-F1 matched accuracy in both cases, the improvement from Random Forest can be attributed to its ability to capture non-linear interactions between features rather than to any class imbalance advantage. For example, the combination of high humidity, high rainfall, and near-neutral pH that jointly separates rice from jute cannot be expressed as a single linear threshold, but tree-based splits capture such conjunctive conditions naturally.

The 5-fold cross-validation result of $99.50\% \pm 0.27\%$ provides further evidence that Random Forest generalizes reliably. The small standard deviation across folds indicates the model's performance is insensitive to the particular train-test partition, and the cross-validation mean slightly exceeding the hold-out accuracy (99.50% vs. 99.32%) is expected, since cross-validation averages over more training data per fold. Taken together, the stability across folds and the agreement between hold-out and CV performance confirm the absence of overfitting, which is a common concern when models reach very high accuracy.

4.2 Feature Interpretation

The feature importance ranking produced by Random Forest provides agronomically meaningful insight into which environmental and soil variables drive crop suitability. Rainfall (0.222) and humidity (0.212) together account for approximately 43% of total importance, establishing water availability as the primary determinant of crop selection in this dataset. This finding aligns with agronomic principles, since rainfall and humidity are largely uncontrollable at the farm level and therefore act as hard environmental filters that eliminate crops whose water requirements cannot be met. The prominence of these two features also explains why the model achieves such high accuracy, because crops are strongly clustered along the water-availability axis, and a single dominant dimension makes the classification problem substantially easier than if all features contributed equally.

Among soil nutrients, potassium (K) ranked highest at 0.179, followed by phosphorus (P) at 0.154 and nitrogen (N) at 0.104. The prominence of potassium is notable, since K is often treated as a secondary nutrient in agronomic discussions relative to N and P, yet in this dataset it is the most discriminative soil variable. This may reflect the role of K in fruit development, drought tolerance, and disease resistance, all of which vary substantially across the 22 crop

classes. Phosphorus, essential for root development and flowering, ranks second, while nitrogen, despite driving leafy vegetative growth, ranks lowest among the three, plausibly because N requirements overlap more broadly across crops than P or K requirements do. Temperature (0.076) and pH (0.054) were the least important features. The low importance of pH is somewhat surprising given how frequently pH is cited in agronomic guidance, but it is consistent with the observation that most crops in this dataset tolerate a relatively narrow and overlapping pH band (roughly 5.5 to 7.5), leaving little room for pH to discriminate between classes.

Taken together, the importance ranking supports a two-stage interpretation of crop suitability: climate (rainfall and humidity) acts as the first filter, determining which crops are viable in a given region at all; soil nutrition (N-P-K) then acts as the second filter, distinguishing between the crops that remain viable. Temperature plays a supporting role in refining recommendations, while pH, in this dataset, contributes only marginally. This two-stage structure has practical implications for deployment. In a production system, rainfall and humidity data — which are readily available from meteorological sources — could be used for a first-pass screening of candidate crops, with soil test results then used to narrow the recommendation to a small top-3 set. A caveat must be noted, however: Random Forest feature importances are not directional. They indicate which features matter most, but not whether higher or lower values of a feature favor a particular crop. Directional interpretation would require SHAP values or permutation-based approaches, which is identified as a direction for future work.

4.3 Limitations

This project has several limitations that should be acknowledged. From a data perspective, while the dataset isn't a completely synthetic data, it was built by augmenting real rainfall, climate, and fertilizer records from India. It is noted that it is exceptionally clean, with no missing values, no duplicates, and perfectly balanced classes. This usually isn't how real-world agricultural data behaves. It won't be this tidy. Instead, it will usually have sensor failures or inconsistent way of keeping records and missing measurements that would've made the data less cleaner. Due to this difference compared to the reality, the model isn't stress-tested against this kind of noise that it would've encountered. As the dataset is relatively small and only covers Indian agronomic conditions, it might not generalize well to other regions with their own soil types, climate zones, etc. Furthermore, seasonal dynamics like changing rainfall patterns or temperatures across a growing season can't be captured as its features are not a time-series data. From a modeling perspective the near-perfect accuracy (99.3% test accuracy, 99.5% CV accuracy) is itself a concern. While cross-validation suggests the model is stable, such high performance on a small, clean dataset raises the possibility of mild overfitting to the dataset's specific distribution rather than genuine generalization to unseen agricultural conditions. The very few misclassifications that did occur were between agronomically similar crops, suggests the model struggles most when feature profiles are genuinely ambiguous. Random Forest outputs class probabilities, but these are often poorly calibrated, meaning the model's stated confidence

may not reflect its true accuracy. Probability calibration (e.g., Platt scaling or isotonic regression) would be required to produce reliable confidence scores for decision-making.

From a scope perspective, this project should be understood as a proof-of-concept rather than a deployable agricultural decision-support system. It demonstrates that crop suitability can be predicted from a small set of soil and climate features with high accuracy, but it does not account for critical real-world factors that also influence much such as market prices, labor availability, crop rotation history, pest and disease risk, irrigation infrastructure, or government agricultural policies. A model that recommends rice based purely on rainfall and humidity, for example, may be agronomically correct but economically unwise if the market price for rice is low that season. Real-world deployment would require integrating these additional constraints, which are absent from the current dataset.

4.4 Future Scope

Several directions could substantially improve this work. The most impactful would be to enrich the dataset with additional features that are able to capture real-world complexity, which can be obtained from integrating real-time IoT soil sensor data (continuously measuring N, P, K, moisture, and temperature), satellite imagery (for vegetation indices like NDVI and land-use classification), historical yield records, weather forecast data, and market price signals. Combining these with the current feature set would allow the model to widen its scope from just choosing which is agronomically suitable but also those who can create more profit for farmers and sustainable for a specific farm that adapts to the current season.

On the modeling side, several alternatives to Random Forest can be worth exploring. XGBoost and LightGBM often outperform Random Forest on tabular data and offer better handling of class imbalance and missing values, which would be valuable if the dataset is later expanded with messier real-world data. Neural networks (particularly multi-layer perceptrons or tabular transformers) could capture more complex non-linear interactions between features, though they would require a much larger dataset to avoid overfitting. Ensemble stacking that combines Logistic Regression, KNN, Decision Tree, and gradient boosting into a single meta-model, could also improve robustness. Finally, adding probability calibration (e.g., Platt scaling or isotonic regression) would allow the model to output meaningful confidence scores, enabling it to flag low-confidence predictions for human review.

Deploying a mobile application with a simple and localized interface that supports different languages and voice inputs for the farmers will be very useful. It will integrate live data, predictions are updated automatically, sensors can be planted so data input are entered directly along with a detailed breakdown of the most prominent factors that drive the prediction so the farmers can gain a deeper understanding from the analysis, as long as making necessary judgements to the model's prediction. To completely transform it into a tool with a wide range that also covers economic aspects, market price data or crop rotation constraints can be integrated as well to support not just maximization of crop production but also the long-term economic decision-making.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

This project set out to build a foundational machine learning classifier that predicts the most suitable crop category from essential soil and environmental features that includes nitrogen, phosphorus, potassium, temperature, humidity, soil pH, and rainfall. Using the publicly available Crop Recommendation dataset, two foundational models were trained and compared using Logistic Regression and Random Forest. Both models achieved high performance on the hold-out test set, with accuracy and F1-scores in the 95–99% range. Random Forest was selected as the final model for its combination of strong predictive performance, stable cross-validated results, and the interpretability offered by its feature importance scores. This interpretability aligns with the growing emphasis on Explainable AI in agricultural machine learning. The feature importance analysis confirmed what agronomic theory predicts: rainfall, humidity, and potassium are the most influential features for distinguishing crop categories, while temperature and pH contribute more subtly. The confusion matrix and error analysis showed that misclassifications are rare and tend to occur between crops with similar soil and climate requirements, which became a reassuring sign that the model has learned meaningful patterns rather than noise.

However, several important limitations must be acknowledged. The dataset's clean structure and perfectly balanced classes overstate real-world model performance. The high accuracy reported here cannot capture the expected complexity of real agricultural decision-making. The dataset also contains no temporal or geographic context, no missing values, and no overlapping suitability ranges, where all of which would be present in real soil and climate data. Additionally, the task is framed as multi-class crop recommendation rather than strict binary suitability classification, due to the one-label-per-record structure of the dataset. Despite these limitations, the project successfully demonstrates a complete, reproducible, zero-cost machine learning pipeline from the process of data inspection and exploratory analysis, through preprocessing and model comparison, to evaluation, interpretation, and deployment via a lightweight Streamlit application. It serves as a solid proof of concept for how foundational classifiers can support crop selection decisions, while remaining honest about what curated data can and cannot tell us.

5.2 Recommendation

Several recommendations emerge from this study, both for improving the technical approach and for moving the work closer to real-world deployment. Future work should focus on three primary directions. First, validating the current pipeline on real agro-climatic data is essential. The dataset used here is clean, balanced, and free of missing values, which is unrepresentative of real farm-level data. Field data typically contains sensor errors, missing measurements, and outliers arising from faulty equipment or human entry mistakes, and it varies

geographically across soil types, elevation, and microclimates. Testing the model on such data would reveal whether the 99.3% accuracy observed here is robust or partially an artifact of the dataset's cleanliness, and would inform the design of preprocessing steps such as imputation, outlier capping, and geographic normalization. Second, the feature set should be extended with seasonal, soil-type, and remote-sensing inputs. The current model relies on seven static features and ignores temporal dynamics such as planting season, monsoon timing, and crop rotation history. Adding these, along with categorical soil-type information and satellite-derived vegetation indices (NDVI, EVI), could improve predictions for borderline cases such as the confusions that are observed in the error analysis. Third, exploring ensemble methods and threshold tuning would improve robustness. While a single Random Forest already performs strongly, additional techniques such as stacking, soft voting, and confidence threshold tuning could reduce residual misclassifications and, more importantly, enable the model to abstain from prediction when confidence is low, which will become a safer behavior for a decision-support tool than forcing a recommendation under uncertainty.

Beyond these technical directions, deployment considerations deserve attention. A production-grade version of this pipeline would benefit from a top-3 recommendation interface rather than a single predicted label, so that farmers and agronomists can compare alternatives that the model considers nearly equally suitable. Coupling predictions with confidence scores, and surfacing interpretable feature contributions (for example through SHAP), would further strengthen user trust and allow domain experts to sanity-check outputs against local knowledge. Integration with existing agricultural advisory platforms and mobile applications would also improve accessibility in regions where farmers primarily rely on smartphones rather than desktop systems. With these improvements, it will become closer to becoming a truly useful decision-support tool for farmers and agronomists, rather than a purely academic exercise. Finally, periodic retraining with up-to-date local data would be necessary to keep recommendations aligned with shifting climate patterns and evolving agricultural practices, ensuring that the system remains useful beyond its initial deployment.

REFERENCES

- A. Ghilan, Y. E. Afou and N. E. Akkad. (2024). Region-Wise Weather Data-Based Crop Recommendation Using Machine Learning Algorithms. IEEE Xplore. <https://xplore.staging.ieee.org/document/10913616>
- Hulatt, L. (n.d.). Crop Selection Strategies. StudySmarter. <https://www.studysmarter.co.uk/explanations/environmental-science/agriculture-and-forestry/crop-selection-strategies/>
- Ahmed, R.A., El-Shafai, W., Ahmed, Z.A. *et al.* (2025). High-precision crop recommendation system with stacking ensemble classifiers for optimizing agricultural productivity. *Sci Rep* 15, 43377 (2025). <https://www.nature.com/articles/s41598-025-09640-5#about-the-journal%232>
- Baldé, B. (2023). Why you should use stratified split. Medium. <https://medium.com/@becaye-balde/why-you-should-use-stratified-split-bddb6dadd34e>
- Bhalla, D. (n.d.) Right Skewed Histogram: Interpretation (with Examples). Listen Data. <https://www.listendata.com/2023/08/right-skewed-histogram.html>
- Clark, B. & Lee, F. (2025). What is Gradient Boosting? IBM. <https://www.ibm.com/think/topics/gradient-boosting>
- Decision Tree in Machine Learning. (2026). GeeksforGeeks. <https://www.geeksforgeeks.org/machine-learning/decision-tree-introduction-example/>
- Food and Agriculture Organization. (2025). The state of the world's land and water resources for food and agriculture 2025: Food supply chains transformation. <https://openknowledge.fao.org/server/api/core/bitstreams/a7f92713-890a-4ff8-85ae-eb2aa76ae702/content/state-of-the-worlds-land-and-water-resources-for-food-and-agriculture-2025-2025/food-supply-chains-transformation.html>
- K-Nearest Neighbor(KNN) Algorithm. (2026). GeeksforGeeks. <https://www.geeksforgeeks.org/machine-learning/k-nearest-neighbours/>
- Kongne Nde, R., Ebongue Kedieng Fendji, J. L., Yenke, B. O., & Schöning, J. (2024). Crop selection. *Smart Agricultural Technology*, 9, 100602. <https://www.sciencedirect.com/science/article/pii/S2772375524002077?via%3Dihub>

- Poornima, S., Mishra, P., Mahishi, R.R. *et al.* (2026). Optimizing crop selection for sustainable agriculture: a compound ensemble approach integrating machine learning and IoT-based sensors. *Sci Rep* 16, 11350 <https://doi.org/10.1038/s41598-026-40772-4>
- Rafi, Md. Abid Hasan, Haque, Md. Dulal, Islam, Nahid, Arman, Md. Sarwar Ali, Karim, Md. Razaul. (2026). Explainable AI-Integrated Real-Time Portable Crop Recommendation System Based on Internet of Things for Precision Agriculture, *Advances in Agriculture*, 1219945, 28 pages, 2026. <https://doi.org/10.1155/aia/1219945>
- Rani, S., Mishra, A.K., Kataria, A. *et al.* Machine learning-based optimal crop selection system in smart agriculture. *Sci Rep* **13**, 15997 (2023). <https://doi.org/10.1038/s41598-023-42356-y>
- R. Grover, R. Grover and S. Goel. (2025). Machine Learning Based Smart Crop Recommendation System. IEEE Xplore. <https://xplore.staging.ieee.org/document/11337597>
- Sharafat, M. S., Kabya, N. D., Emu, R. I., Ahmed, M. U., Onik, J. C., Islam, M. A., & Khan, R. (2025). An IoT-enabled AI system for real-time crop prediction using soil and weather data in precision agriculture. *Smart Agricultural Technology*, 12, Article 101263. <https://www.sciencedirect.com/science/article/pii/S2772375525004940#1>
- Singhal, S., Suthar, O. P., & Mishra, A. (2025). State-of-the-art and emerging trends in crop recommendation systems: A comprehensive review. *Procedia Computer Science*. <https://www.sciencedirect.com/science/article/pii/S1877050925010907>
- S. Nimmala, V. Bindla, C. Veena, G. Sirisha, G. Sampath and K. Pabba. (2025). Fuzzy-Enhanced Apriori for Crop Recommendation Based on SE Parameters. IEEE Xplore. <https://ieeexplore.ieee.org/document/11256313#1>
- Soil pH. (n.d.). Mosaic. <https://www.croptonutrition.com/nutrient-management/soil-ph/>